INCH-POUND

MIL-M-28787/71C(SH)
AMENDMENT 4
29 March 1991
SUPERSEDING
AMENDMENT 3
16 January 1986

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SHIFT REGISTER, DIGITAL, KEY CODE CHF

This amendment forms a part of Military Specification MIL-M-28787/71C(EC), dated 9 September 1982, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

TABLE I, Test conditions H1-H16, Max columns: Delete "500" (four places) and substitute "400" (four places).

* TABLE I, Test conditions D1-D16 and G1-G22, Test type column: Delete "X" and substitute "S".

* TABLE I, Test conditions L1-L18, Max columns: Delete "-430" (four places) and substitute "-510" (four places).

* TABLE I, Test conditions L19-L21, Max columns: Delete "-1.58" (four places) and substitute "-1.94" (four places).

PAGE 7

* 4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through AA shall be performed in any order with the following exceptions. Test conditions A, B, C, D and G are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, D or G are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, and G where applicable, followed by the remaining tests in the sequence."

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

* 4.5.1: Delete and substitute the following:

"4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through AA in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E, F, H through Y, and AA shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2."

* 4.5.3: Delete and substitute the following:

"4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests E, F, H through Y, and AA shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests E, F, H through Y, and AA shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I."

6.4, Maximum power dissipation: Delete "440" and substitute "506".

PAGE 8

6.4, Input current, P1-P16, R, L: Low level (V_{IL} = 0.0V): Delete "-0.43" and substitute "-0.51".

6.4, Input current, P1-P16, R, L: Low level (V_{IL} = 0.4V): Delete "-0.36" and substitute " -0.44".

6.4, Input current, S0, S1, CLR: Low level (V_{IL} = 0.0V): Delete "-1.58" and substitute "-1.94".

6.4, Input current, S0, S1, CLR: Low level (V_{IL} = 0.4V): Delete "-1.44" and substitute "-1.76".

6.4, Output voltage, Low level: Delete "6" and substitute "4" and delete "0.5" and substitute "0.4".

6.4, Output drive current maximum capability, Low level: Delete "0.5" and substitute "0.4" and delete "6" and substitute "4".

PAGE 44

TABLE X, Test conditions H1-H16, Max limits column: Delete "500" and substitute "400".

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

PAGE 50

FIGURE 17, NOTE 2: Delete "6" and substitute "4" and delete "0.5" and substitute "0.4".

PAGE 54

* TABLE XIII, Test condition L1, Max column: Delete "-430" (one place) and substitute "-510" (one place).

* TABLE XIII, Test condition L19, Max column: Delete "-1.58" (one place) and substitute "-1.94" (one place).

PAGE 55

* TABLE XIII, Test condition L20, Max column: Delete "-1.58" (one place) and substitute "-1.94" (one place).

* TABLE XIII, Test condition L21, Max column: Delete "-1.58" (one place) and substitute "-1.94" (one place).

PAGE 66

* TABLE XVII, Test condition Q17, B to pin column: Add "6" between "3" and "7".

PAGE 110

* TABLE XXVI, Test conditions AA49 to AA64, Max limit column: Delete "1.5" and substitute "700".

* TABLE XXVI, Test conditions AA49 to AA64, Unit column: Delete "V" and substitute "mV".

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

FIGURE 41, NOTES: Delete notes 3 and 4 and add the following new tests:

"3. Item J01-3 pulse parameters for test conditions AA1-AA16:
PRP3 = 1000 ± 100ns.
PWC = 25 ± 2ns.
$t_{TLH3} = T_{THL3} = 6 \pm 2$ns.

4. Item J01-3 pulse parameters for test conditions AA17-AA48:
PRP3 = 1000 ± 100ns.
PWC = 30 ± 2ns.
$t_{TLH3} = T_{THL3} = 6 \pm 2$ns.

5. Sweep the item J01-3 pulse through one cycle. The output pulse at point B shall meet the requirements of table XXVI (test conditions AA1-AA48)."

Following figure 41, add new figure 42 as printed on page 5 of this amendment.

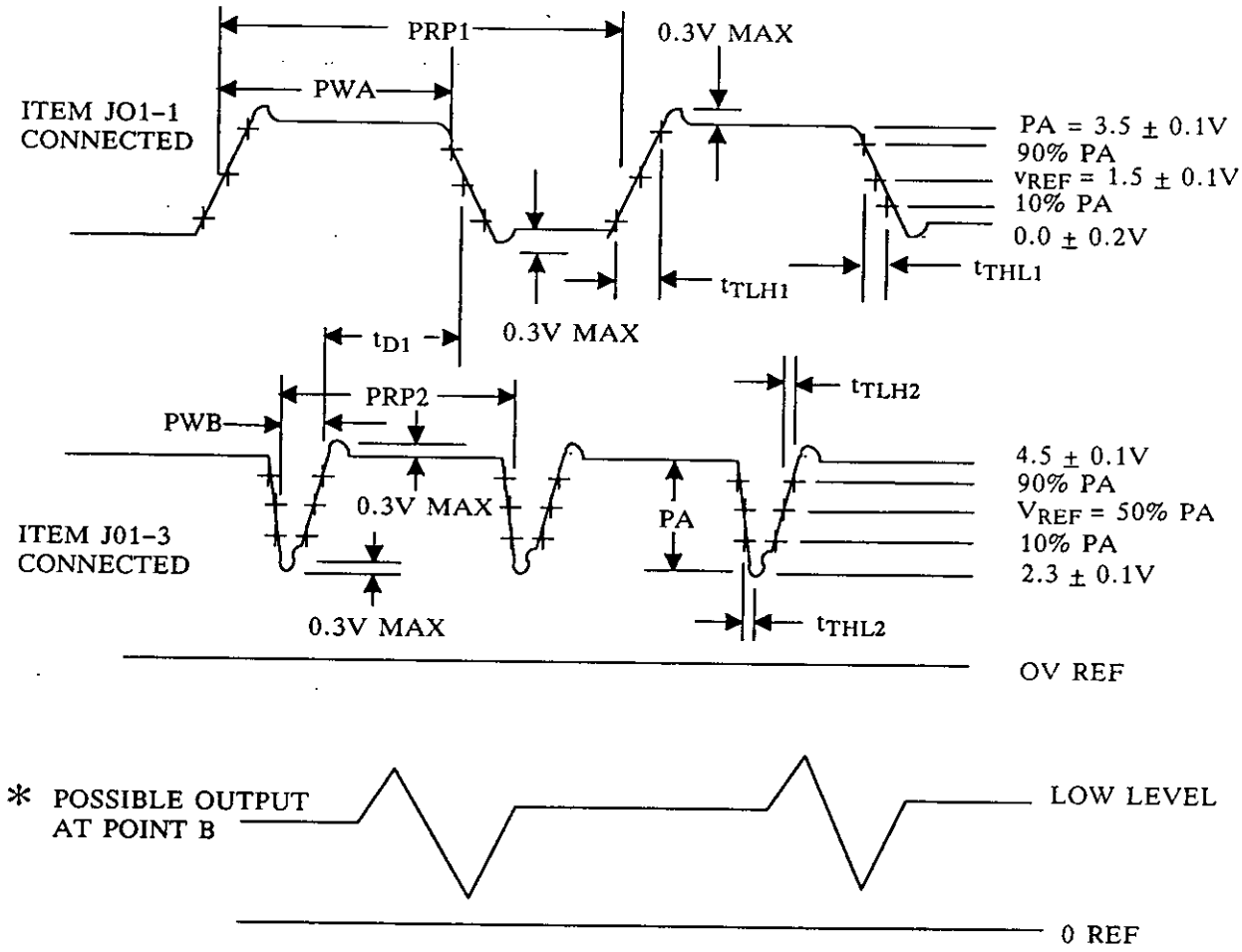
NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
Navy - SH

(Project 5963-N771)

Agent: NW

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. Item J01-1 pulse parameters:
   PRP1 = 1000 ± 100ns.
   PWA = 500 ± 50ns.
   $t_{TLH1} = t_{THL1} =$ 10 ± 2ns.
   t_{D1} = 100 ± 10ns.
2. Item J01-3 pulse parameters:
   PRP2 = 500 ± 100ns.
   PWB = 20 ± 5ns.
   $t_{TLH2} = t_{THL2}$ = 6 ± 2ns.
3. The output pulse at point B shall meet the requirements of table XXVI (test conditions AA49-AA64).

FIGURE 42. Noise sensitivity test circuit waveforms for table XXVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.